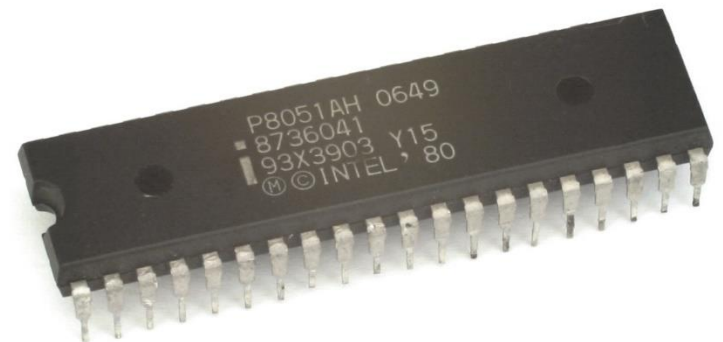


Microprocessor Systems

Lab4 – 1 Timer and Interrupt Service Routine

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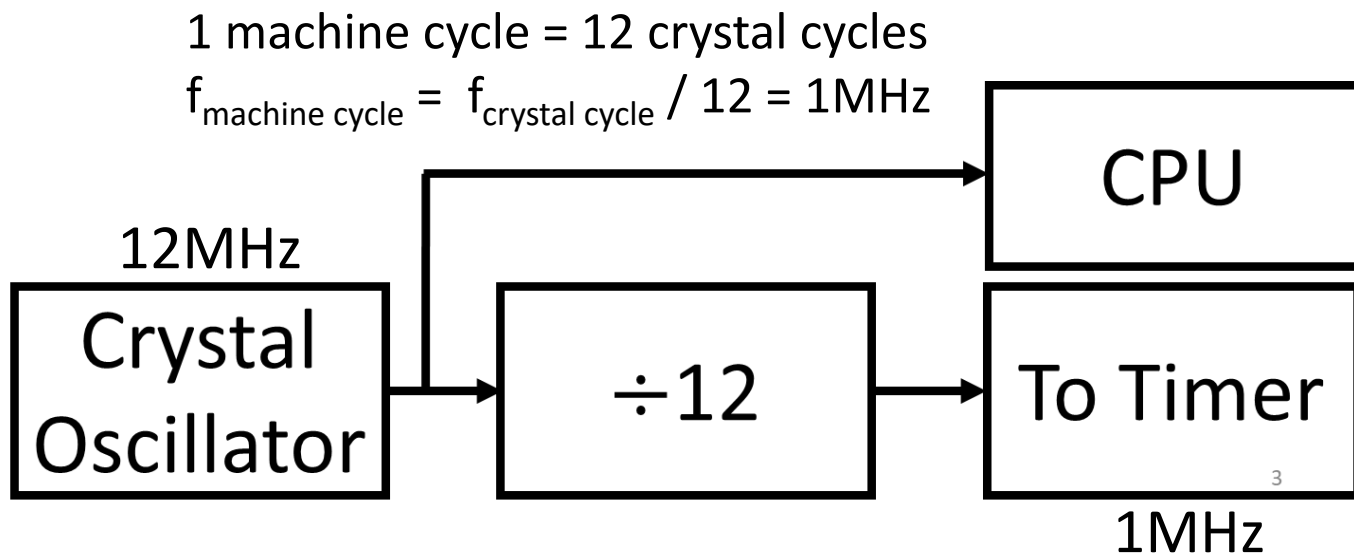
Introduction

- 8051 provides two timers/counters which enable
 - Counting external events
 - Timer interrupt service routine
- The two timers, Timer 0 and Timer 1, are each composed of two 8-bit registers.
 - TH0, TL0: Registers of higher and lower 8 bits of Timer 0.
 - TH1, TL1: Registers of higher and lower 8 bits of Timer 1.

Timer	Timer 0																Timer 1															
Reg. Name	TH0								TL0								TH1								TL1							
Bit NO.	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0

Clock

- 8051 takes an external crystal as the clock source of its timers. (1/12 of the crystal frequency)
- An assembly instruction takes at least 12 crystal cycles to be executed, which means CPU machine cycle frequency is also 1/12 of the crystal frequency.



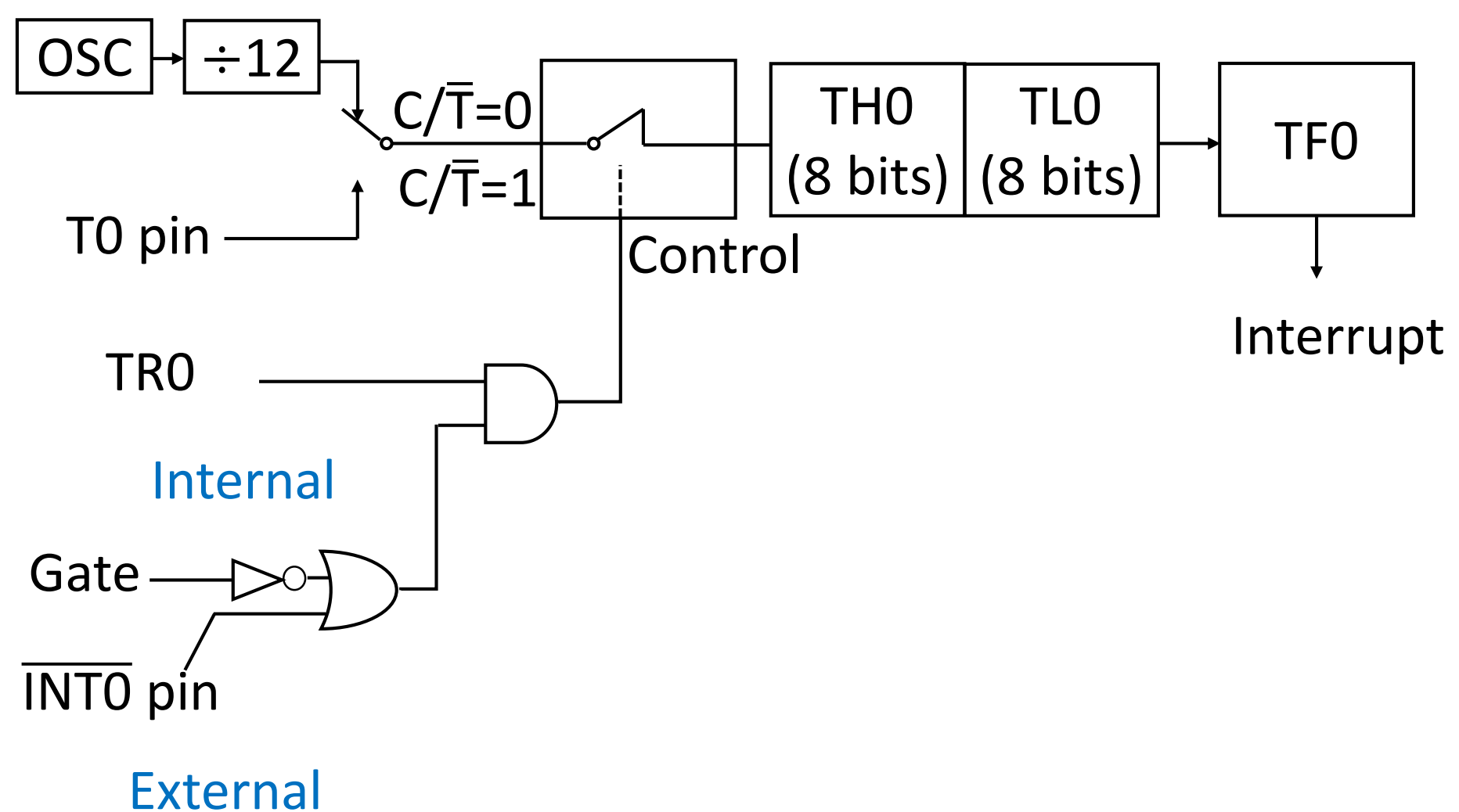
Time Mode Control Register (TMOD)

Mode Selection

	Timer1				Timer0			
Bit No.	7	6	5	4	3	2	1	0
Symbol	Gate	$C/\bar{T}$	M1	M0	Gate	$C/\bar{T}$	M1	M0

M1	M0	Mode	Description
0	0	0	13-bit counter
0	1	1	16-bit counter
1	0	2	8-bit counter with autoreload
1	1	3	Split Timer0 into 8-bit counters or stop Timer1

Mode 1 (16-bit Counter)



Time Mode Control Register (TMOD)

Enable and Clock Source Selection

	Timer1				Timer0			
Bit No.	7	6	5	4	3	2	1	0
Symbol	Gate	$C/\overline{T}$	M1	M0	Gate	$C/\overline{T}$	M1	M0

	Description
Gate = 0	Timer is enabled by setting TRX = 1. (X = 0 or 1)
Gate = 1	Timer is enabled by setting both TRX = 1 and INTX = 1. (X = 0 or 1)
$C/\overline{T} = 0$	Act as a timer, which counts the number of machine cycles. (Internal clock source)
$C/\overline{T} = 1$	Act as a counter, which counts negative transitions at pinT0 or T1. (External clock source)

Time Control Register (TCON)

- TF0/1: Timer0/1 overflow flag
 - Set by hardware when the timer overflows.
 - Cleared by hardware when the interrupt routine is served.
- TR0/1: Timer0/1 run control bit
 - Set/Cleared by software to turn the timer on/off.

Bit No.	7	6	5	4	3	2	1	0
Symbol	TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0

Time Control Register (TCON)

- IE0/1: Timer0/1 interrupt enable flag
 - Set by hardware when the interruption of INT0/1 is detected.
 - Cleared by hardware when the interrupt routine is served.
- IT0/1: Timer0/1 interrupt type
 - Set to 1 means negative edge triggered.
 - Set to 0 means low level triggered.

Bit No.	7	6	5	4	3	2	1	0
Symbol	TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0

IE (Interrupt Enable) Register

EA	-	-	ES	ET1	EX1	ET0	EX0
----	---	---	----	-----	-----	-----	-----

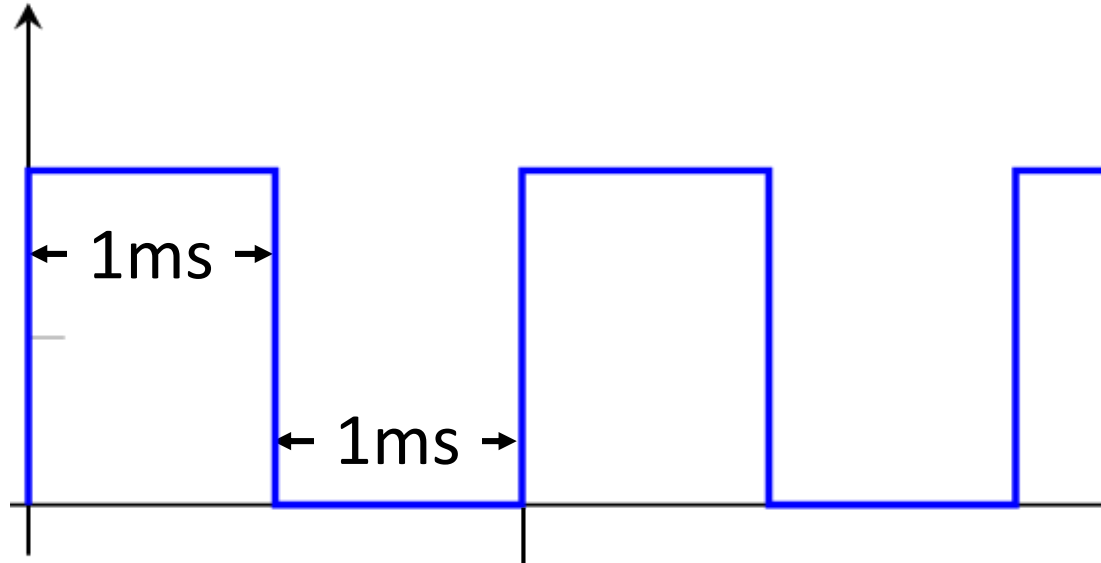
EA	Enables/disables all interrupt
-	Reserved for future use
-	Reserved for future use
ES	Enables/disables serial port interrupt
ET1	Enables/disables timer1 overflow interrupt
EX1	Enables/disables external interrupt1
ET0	Enables/disables timer0 overflow interrupt
EX0	Enables/disables external interrupt0

Interrupt Priority

Interrupt Source	Vector address	Interrupt priority
External Interrupt 0 –INT0	0003H	0
Timer 0 Interrupt	000BH	1
External Interrupt 1 –INT1	0013H	2
Timer 1 Interrupt	001BH	3
Serial Interrupt	0023H	4

Generate a square wave of 2ms

- Assume crystal oscillator frequency of 12MHz
 - one timer cycle = $1\mu\text{s}$
 - $\text{count} = (1\text{ms}) / (1\mu\text{s}) = (1000\mu\text{s}) / (1\mu\text{s}) = 1000$



Generate a square wave of 2ms

- Take Timer0 mode1 as an example, the timer will increment from 0 ~ 65535 .
- To interrupt after counting 1ms, the initial value to be set for the timer will be 1000 less than its maximum.
- i.e., set initial time value to (65535-1000), and map this 16-bit value into two 8-bit registers.

$TH0 = (65536 - 1000) / 256 = 0xFC$ // 8 bit quotient

$TL0 = (65536 - 1000) \% 256 = 0x18$ // 8 bit remainder

Generate a square wave of 2ms

```
1 #include "8051.h"
2
3 int cnt = 0;
4
5 void T0_isr(void) __interrupt(1)
6 {
7     // PC jumps to here every 1ms
8     TH0 = (65536 - 1000) / 256;
9     TH1 = (65536 - 1000) % 256;
10
11     cnt++;
12     if (cnt >= 1000) {
13         cnt = 0;
14         P1 = ~P1;
15     }
16 }
17
18 void main(void)
19 {
20     // Intialization
21     TMOD = 0x01;
22     TH0 = (65536 - 1000) / 256;
23     TH1 = (65536 - 1000) % 256;
24     ET0 = 1;
25     EA = 1;
26     TR0 = 1;
27
28     // Run until power off
29     while (1);
30 }
```

interrupt priority

// Global variable for interrupt routing

// interrupt routine with priority 1

// Reset higher 8 bits of Timer0

// Reset lower 8 bits of Timer0

// Count each interruption

// 1000 interruptions = 1000ms = 1s

// Reset count

// Invert wave signal

// Set Timer1 to mode 0 and Timer0 to mode 1

// Load initial higher 8 bits into Timer0

// Load initial lower 8 bits into Timer0

// Enable Timer0 interrupt

// Enable all interrupt

// Start Timer0

Generate a square wave of 2ms (Polling)

```
1 #include "8051.h"
2
3 int cnt = 0;
4
5 void timer_delay(void)           // Delay function, delay 1ms
6 {
7     TH0 = (65536 - 1000) / 256;  // Load higher 8 bits into Timer0
8     TL0 = (65536 - 1000) % 256;  // Load lower 8 bits into Timer0
9     TR0 = 1;                     // Start Timer0
10    TF0 = 0;                     // Clear Timer0 flag
11    while (THF == 0);             // Wait until Timer0 flag is set (1ms)
12    TR0 = 0;                     // Stop Timer0
13 }
14
15 void main(void)
16 {
17     P1 = 0x00;                  // Initial wave signal
18     TMOD = 0x01;                // Set Timer1 to mode 0 and Timer0 to mode 1
19     while (1) {
20         P1 = ~P1;               // Invert wave signal
21         fct (cnt = 0; cnt < 1000; cnt++) // Do a thousand times
22             timer_delay();      // Repeatedly call timer_delay
23     }
24 }
```

Reference

- <https://www.electronicwings.com/8051/8051-timers>
- <https://www.slideshare.net/ishonp98/8051-timers-and-counters>
- <http://dns2.asia.edu.tw/~rikki/mps107/mcs-ch07.pdf>
- <http://tw.classf0001.uschoolnet.com/css000000183803/cm4k-1358643211-5690-3573.pdf>

Microprocessor Systems

Lab4 – 2 Hall Magnetic Sensor

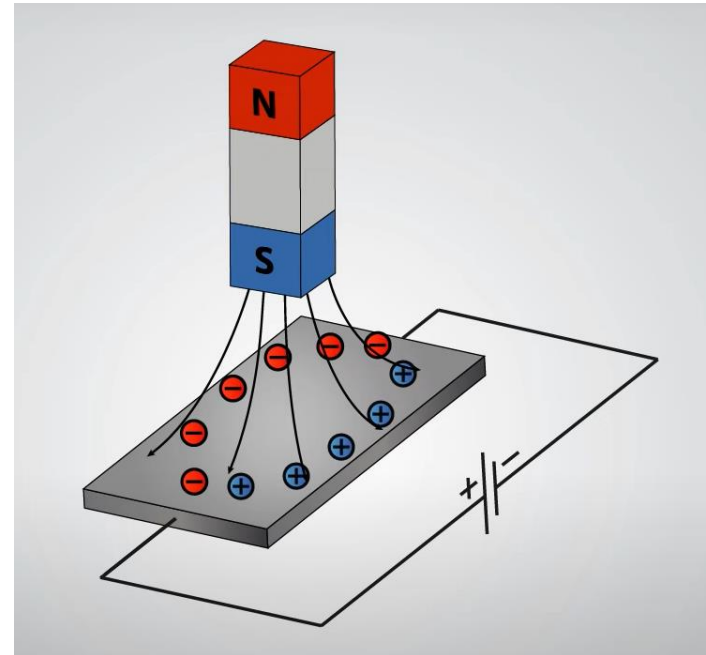
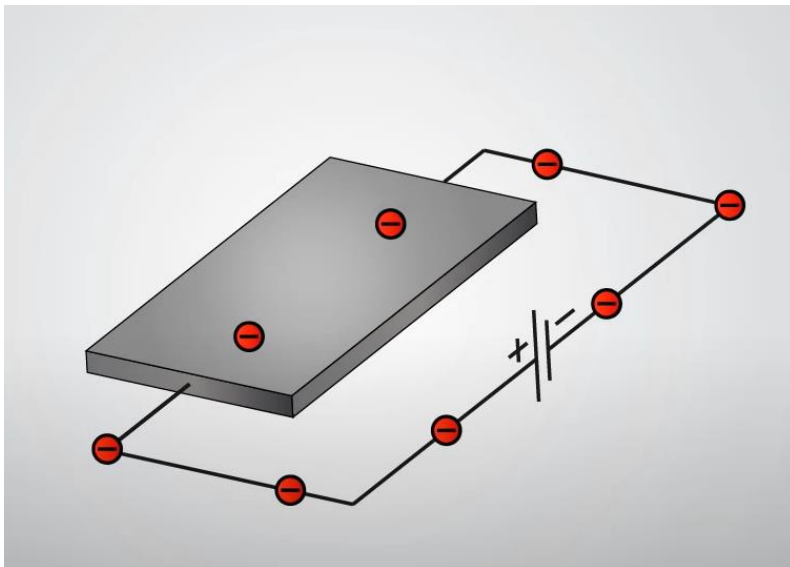
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Hall Effect

- Discovered by Edwin Hall in 1879.
- When magnetic flux goes through a conductive plate, it separates the positive and negative charges, creating Hall voltage.

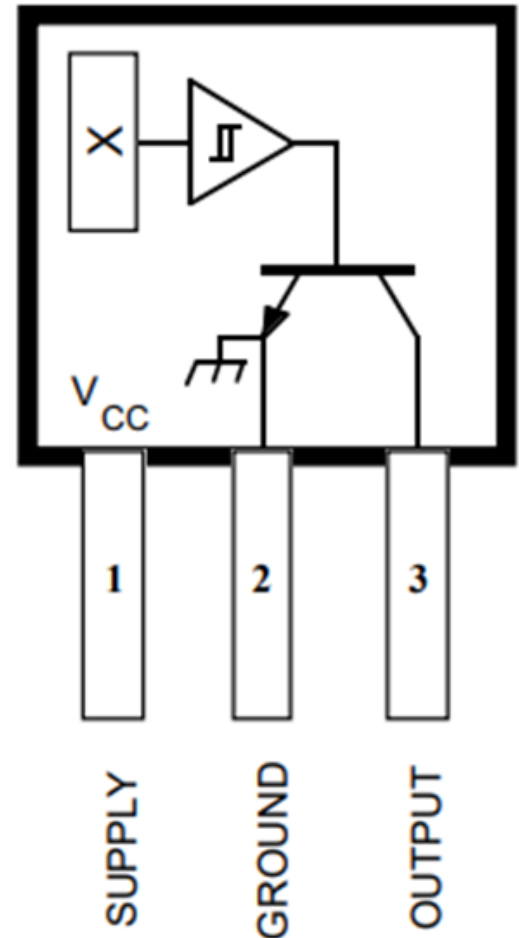
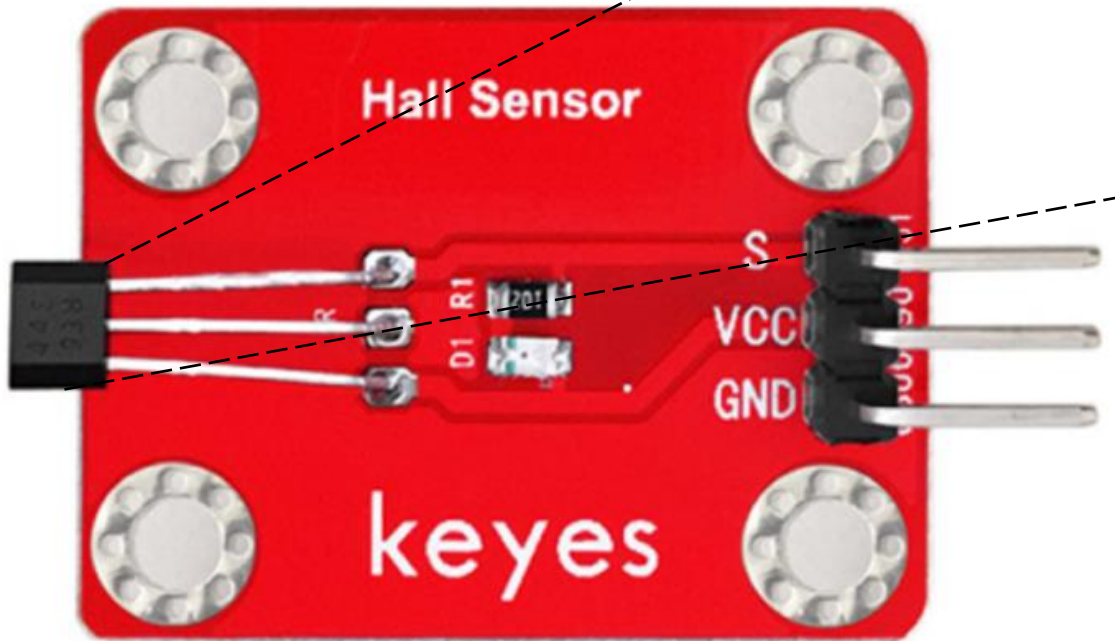


Hall Magnetic Sensor

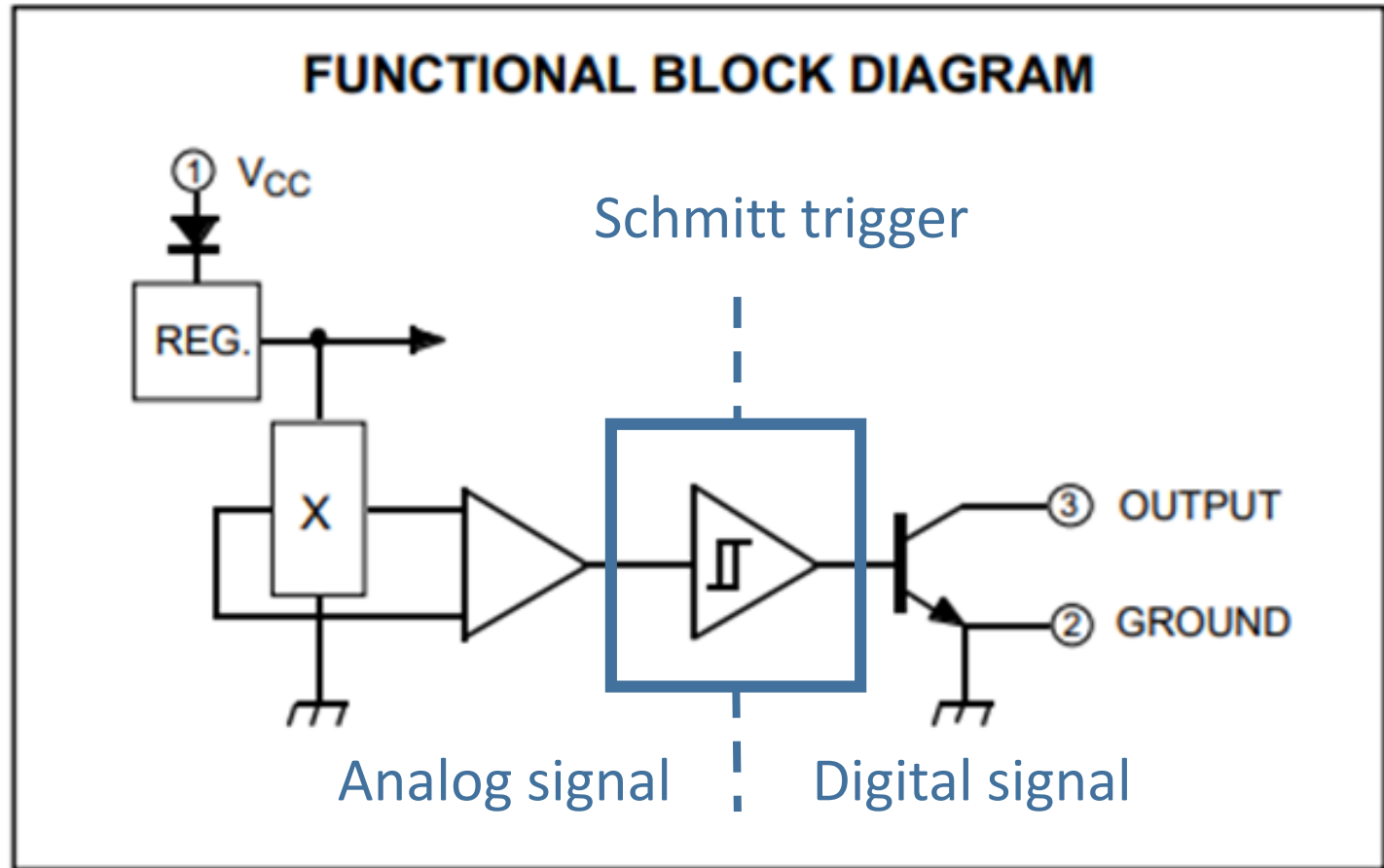
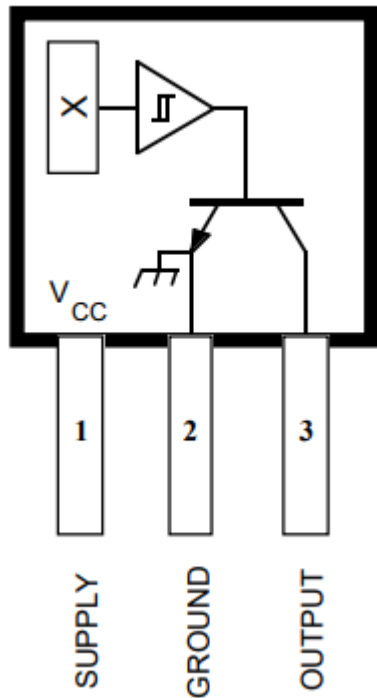
- A Hall magnetic sensor is a device which is activated by external magnetic fields.
- It converts the magnetic flux density around it into output voltage (Hall voltage), which is usually a few micro Volt per Gauss.
- Hall sensors usually consist of a thin piece of rectangular p-type semiconductor material, where the Hall effect takes place.

Hall Magnetic Sensor

- In this course, we use Hall Magnetic Sensor A3144E.

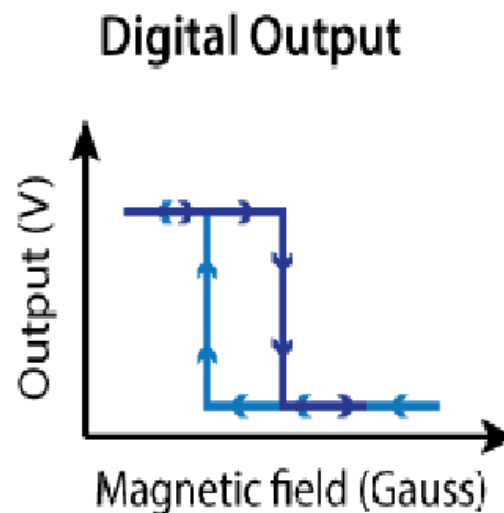
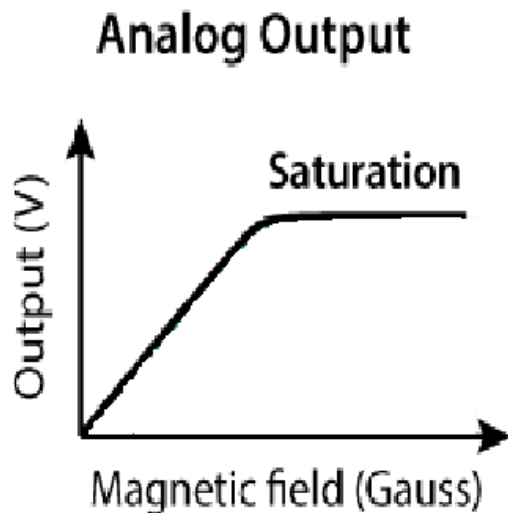


Block Diagram



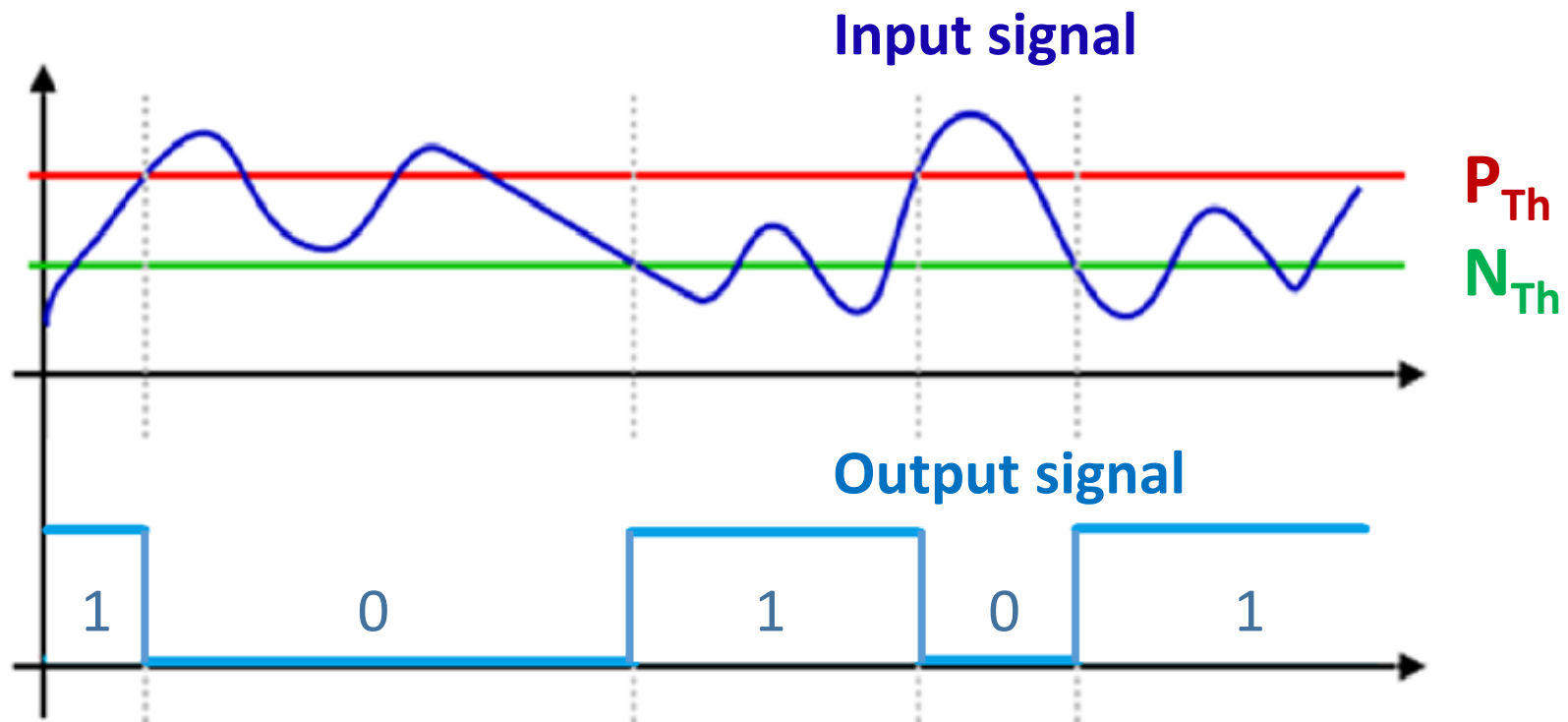
Types of Hall Effect ICs

- **Linear output type:** Used to obtain an output voltage that is directly proportional to the magnetic field strength.
- **Switching type:** Used to obtain an on/off signal.



Schmitt Trigger

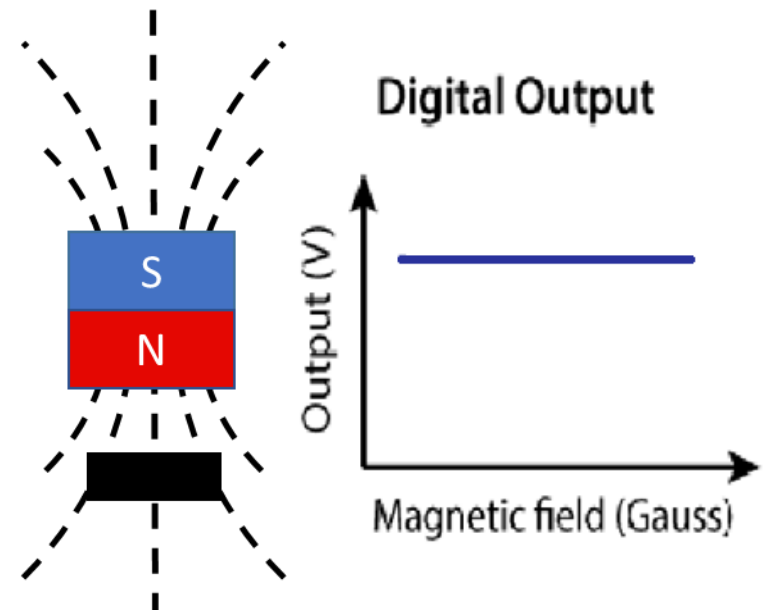
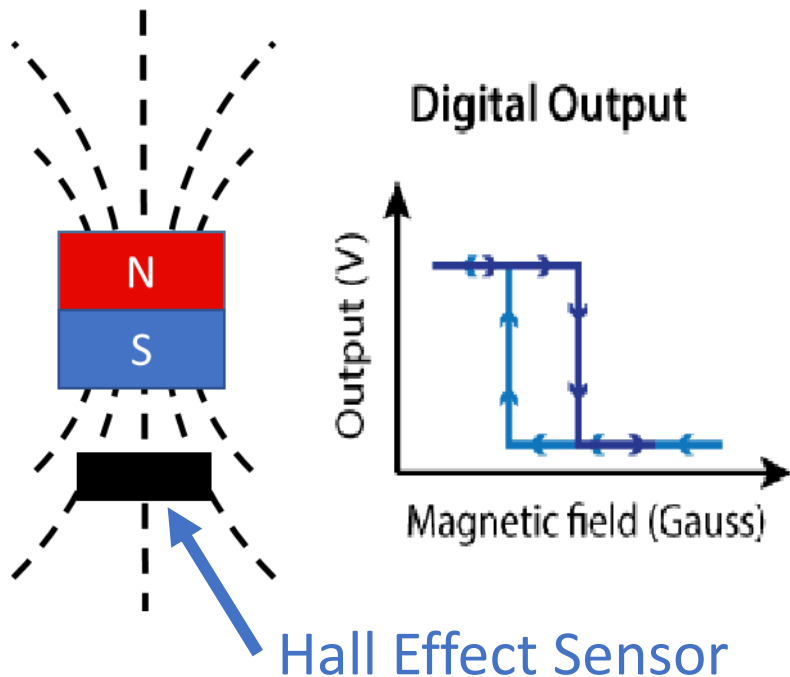
- Input signal $>$ Positive threshold: Output signal = 0
- Input signal $<$ Negative threshold: Output signal = 1



Type of Hall Effect IC Detection

- **Unipolar Detection**

- Only one magnetic field pole (north or south) is detected.
- ON/OFF operation is performed according to the magnetic flux density to output a high or low level signal.

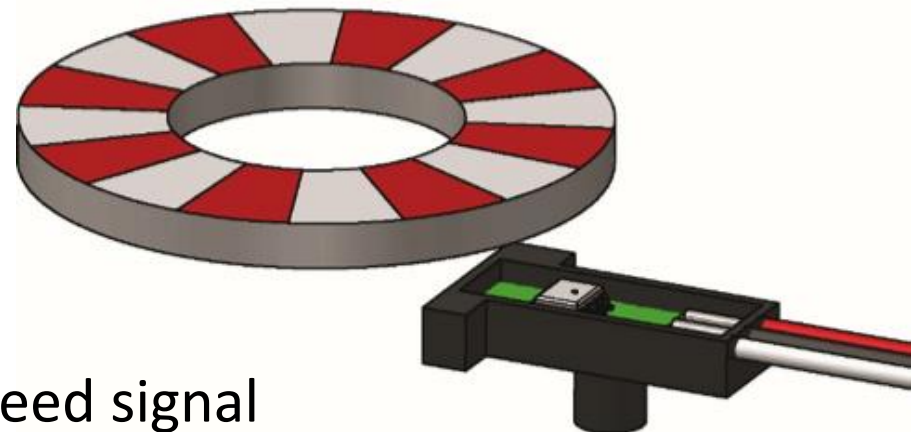
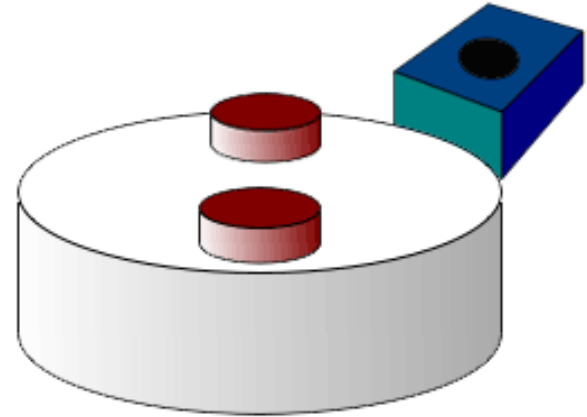


Applications of Hall Effect Sensors

- Proximity sensing
- Positioning
- Speed detection
- Current sensing

A rotating 16-pole ring magnet
Red (north pole) : Activate
White (south pole) : Deactivate

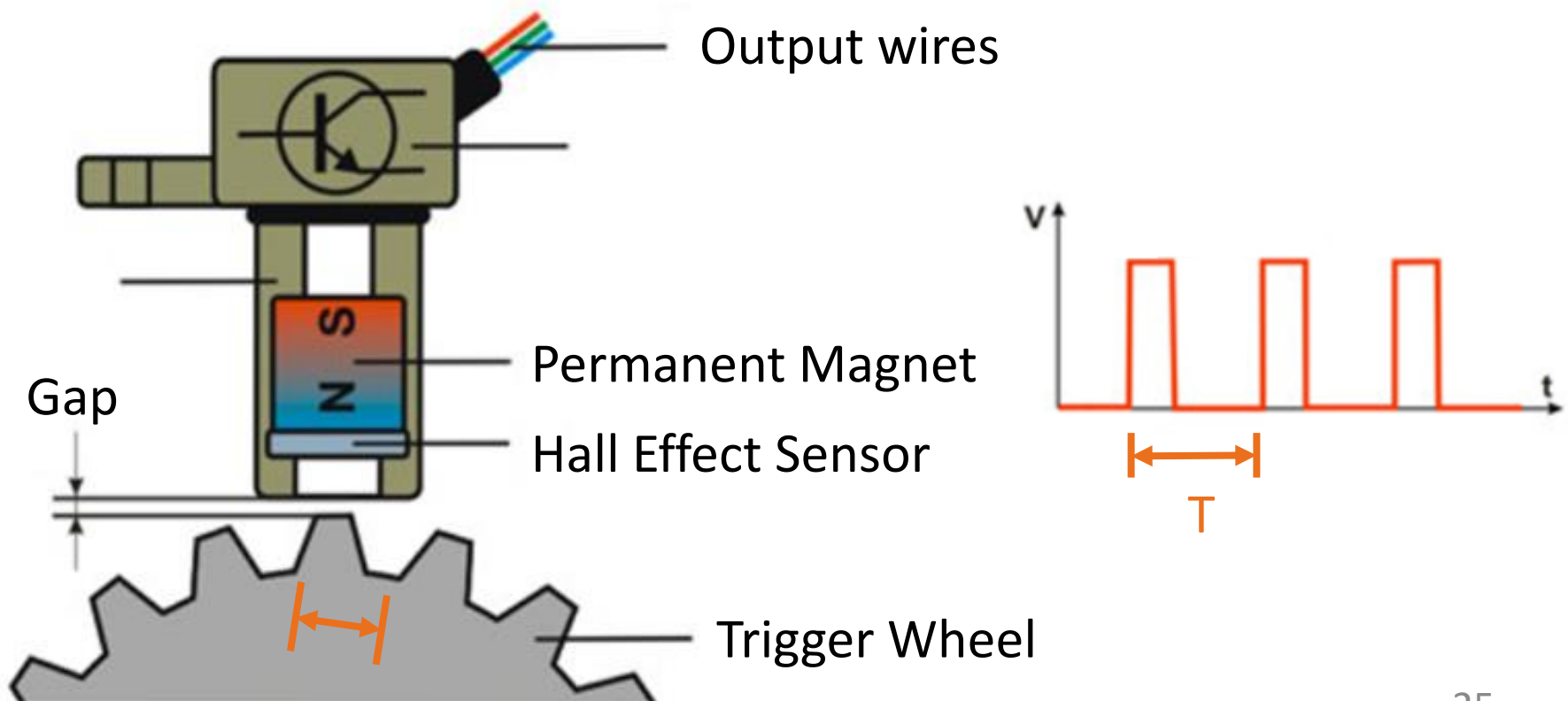
→ resulting in a very accurate speed signal



Hall Effect Speed Sensor

Applications of Hall Effect Sensors

- Wheel speed sensors – RPM
- The frequency increases proportionally to rpm



Reference

- <https://www.allaboutcircuits.com/industry-articles/a-guide-to-the-applications-of-hall-effect-and-reed-switch-sensors/>
- <https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html>
- <https://www.ablic.com/en/semicon/products/sensor/magnetism-sensor-ic/intro/>
- <https://howtomechatronics.com/how-it-works/electrical-engineering/hall-effect-hall-effect-sensors-work/>
- <https://www.linkedin.com/pulse/inductive-hall-effect-rpm-sensors-explained-kiril-mucevski?trk=mp-author-card>